

Requirement Analysis Phase

Data Flow Diagrams (DFD) & Agile User Stories

Attribute	Project Information
Project Title	Auto Ticket Classification using Flow Designer
Phase / Stage	2. Requirement Analysis
Document Title	DFD & Agile User Stories
Platform & Scope	ServiceNow Utah / Vancouver / Washington DC (Global Scope)
Team ID	SWTID-2026-4988

1. Data Flow Diagrams (DFD Levels)

Data flow diagrams illustrate the movement and transformation of incident data as it enters the ServiceNow platform, undergoes keyword evaluation, and triggers downstream actions.

Level 0: Context Data Flow Diagram

At the macro level, the external entity (Caller: Student/Teacher) submits an incident into the ServiceNow System. The system automatically processes the data, classifies the record, updates the internal database, and returns an automated email notification and status confirmation to the Caller.

Level 1: Functional Decomposition DFD

Process ID	Process Name	Input Data Flow	Output Data Flow	Data Store / Target
1.0	Incident Form Capture	Caller Name, Short Description from UI form	Validated raw record payload	Buffer / GlideRecord Object
2.0	Table Record Insertion	Validated raw record payload	Stored record with INC005xx number	Data Store: u_incident_workflow
3.0	Trigger Evaluation	New record with empty Category	Flow Designer Execution Context	ServiceNow Event Engine
4.0	Keyword Parsing & Classification	Short Description string contents	Assigned Category & Subcategory	Flow Logic Engine (If/Else If)
5.0	Record Update Execution	Category & Subcategory values	Updated record state	Data Store: u_incident_workflow
6.0	Email Dispatch & Logging	Caller.email, Subject, Body payload	Dispatched SMTP message & audit log	Data Store: sys_email / Outbox

2. Agile User Stories & Backlog

The project functional scope is broken down into modular Agile User Stories prioritized across development sprints:

Story ID	Epic	User Description	Story	Acceptance Criteria	Story Points	Priority	Sprint
USN-1	Data Architecture	As a ServiceNow Administrator, I want to create a custom table "Incident WorkFlow" (u_incident_workflow) so that school IT tickets are stored in a dedicated schema.		Table exists with label "Incident WorkFlow", technical name "u_incident_workflow", and Global application scope.	3	High	Sprint 1
USN-2	Data Architecture	As a Helpdesk Manager, I want each ticket to have a unique auto-generated number starting with INC00500 so tickets can be tracked cleanly.		Auto-numbering active; prefix "INC", starting number 500, digits 5. Test records produce INC00509, INC00510, etc.	2	High	Sprint 1
USN-3	Data Architecture	As a Support Specialist, I want Category and Subcategory fields with dependent choice logic so only valid subcategories are selectable.		Category choices: Network, Hardware, Access, Performance. Subcategory choices dynamically filter based on selected Category.	3	High	Sprint 1
USN-4	User Experience	As a Student/Teacher, I want to submit a ticket by entering only my name and issue description without navigating confusing IT categories.		Form Designer shows Caller and Short Description prominently; Category field is optional on submission form.	2	High	Sprint 1
USN-5	Flow Automation	As a Network Specialist, I want tickets mentioning "WiFi" or "Network" to automatically be assigned Category=Network and Subcategory=Wi-Fi.		Short description with "WiFi" or "Network" updates record to Category=Network, Subcategory=Wi-Fi upon record creation.	3	High	Sprint 2

Story ID	Epic	User Story Description	Acceptance Criteria	Story Points	Priority	Sprint
USN-6	Flow Automation	As a Hardware Technician, I want tickets mentioning "Projector" to automatically be assigned Category=Hardware and Subcategory=Projector.	Short description with "Projector" updates record to Category=Hardware, Subcategory=Projector.	3	High	Sprint 2
USN-7	Flow Automation	As an Identity Admin, I want tickets mentioning "Forgot password" or "Login" to automatically be assigned Category=Access and Subcategory=Forgot Password.	Short description with "Forgot password" updates record to Category=Access, Subcategory=Forgot Password.	3	High	Sprint 2
USN-8	Flow Automation	As a Desktop Support Tech, I want tickets mentioning "Slow Computer" to automatically be assigned Category=Performance and Subcategory=Slow Computer.	Short description with "Slow Computer" updates record to Category=Performance, Subcategory=Slow Computer.	3	High	Sprint 2
USN-9	Notification	As a Ticket Caller, I want to receive an immediate email confirmation upon ticket submission so I know IT is addressing my issue.	Flow Designer Send Email action sends notification to Caller.email with subject "Your Request for the issue has been submitted."	3	High	Sprint 2
USN-10	Release Management	As a Release Manager, I want all configuration changes captured in an Update Set so they can be exported to XML and promoted.	Local Update Set "Project Update Set" marked Complete and exported to XML without errors.	2	Medium	Sprint 2